Determination of voltage in circuit using Nodal analysis

AIM

To Determine the voltage in the circuit using nodal analysis both theoretically and practically for a given DC circuit

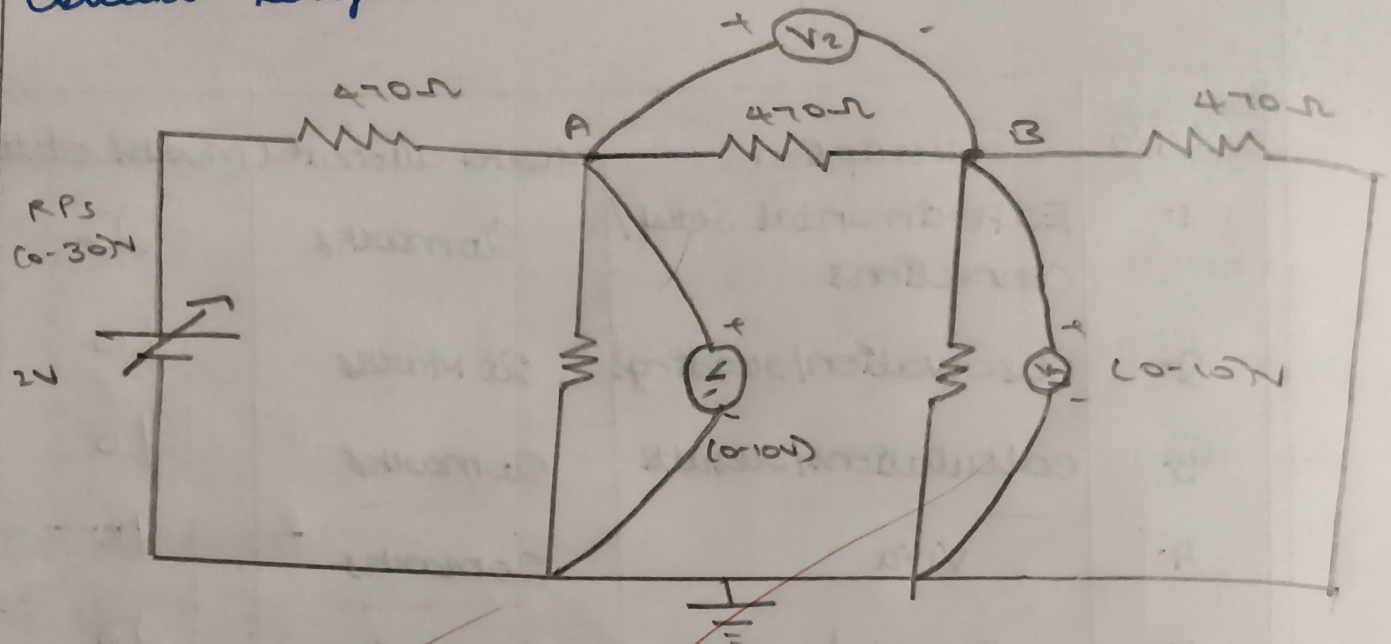
APPARATUS Required

S.No	Apparatus	Specification	Quantity
1.	Regulated Power Supply (RPS)	(0 - 30V)	1
2.	Multimeter	–	1
3.	Resistor	470 Ω, 330 Ω	3.2
4.	Bread Board	–	1

Procedure

* Given connections as per the circuit Diagram
* Switch on the supply, vary the RPS (regulated Power supply) and set a particular input voltage
* Note down the readings of ammeters and voltmeter and tabulate them.
* vary the RPS to its minimum value and switch off the supply.

Circuit Diagram



Tabular column

Parameters	Theoretical	Practical
2V	0.92	0.96
5V	2.61	2.68
10V	5.39	5.47
14V	7.59	7.66

Calculations:

* For 2V

$V_1 = 0.20, V_2 = 0.72$

$V = V_1 + V_2$

$V = 0.92V$

* For 5V

$V_1 = 0.60, V_2 = 2.01$

$V = V_1 + V_2$

$V = 0.60 + 2.01$

$V = 2.61V$

* For 10V

$V_1 = 1.30, V_2 = 4.09$

$V = V_1 + V_2$

$V = 5.39V$

* For 14V

$V_1 = 1.81, V_2 = 5.78$

$V = V_1 + V_2$

$V = 1.81 + 5.78$

~~$V = 7.59V$~~ $V = 7.59V$

* Vary the RPS for different input voltage and note down the readings of all the meters
* Using the tabulated values, verify Kirchoff's laws practically, and verify it theoretically.

Marks Split up

S.No	Description	Marks allowed	Marks obtained
1.)	Experimental Setup/ Connections	20 Marks	20
2.)	Observations / reading	20 Marks	10
3.)	Calculation/Results	20 Marks	10
4.)	Viva	20 Marks	20
5.)	Record	20 Marks	20
			80

80

Result:

Thus the nodal analysis is verified Pratically and theoretically. The resultant voltage for 2 V supply are

a) The voltage V_1 is 0.96 V

b) The voltage V_2 is 2.68 V.

c) The voltage V_3 is 5.47 V

d) The voltage V_4 is 7.66 V.